

ALGEBRA I FULL-YEAR COMPUTATIONAL COURSE

Student Book and Teacher Answer Key

Course design: full-year Algebra I, unit by unit, with computational practice only. This book intentionally avoids word problems and focuses on symbolic, numeric, tabular, and graph-ready algebra skills.

How to Use This Book

Each unit contains:

1. Learning targets aligned to Algebra I topics.
2. Formula sheet for quick reference.
3. Sample problems with worked solutions.
4. Classwork practice for guided or independent class use.
5. Homework practice for additional repetition.
6. Teacher answer key for every classwork and homework item.

YEAR-AT-A-GLANCE PACING

Unit | Topic | Suggested Time

- | | | |
|----|---|---------|
| 1 | Real Numbers, Properties, and Precision | 2 weeks |
| 2 | Exponents, Radicals, and Rational Exponents | 3 weeks |
| 3 | Expressions, Structure, and Equivalent Forms | 3 weeks |
| 4 | Linear Equations, Inequalities, and Absolute Value | 4 weeks |
| 5 | Coordinate Graphing and Linear Functions | 4 weeks |
| 6 | Systems of Equations and Inequalities | 4 weeks |
| 7 | Functions, Notation, Domain, Range, and Transformations | 4 weeks |
| 8 | Polynomials and Factoring | 4 weeks |
| 9 | Quadratic Functions and Equations | 5 weeks |
| 10 | Exponential Functions and Sequences | 3 weeks |
| 11 | Rational Expressions, Radicals, and Inverses | 3 weeks |
| 12 | Statistics, Data Displays, and Linear Models | 3 weeks |
| 13 | Cumulative Review and Final Exam Practice | 2 weeks |

COURSEWIDE FORMULA SHEETS

Number and Quantity

- Natural numbers: 1,2,3,...

- Whole numbers: 0,1,2,3,...
- Integers: ..., -2, -1, 0, 1, 2, ...
- Rational numbers: numbers that can be written as $\frac{a}{b}$, where a,b are integers and $b \neq 0$
- Irrational numbers: non-repeating, non-terminating decimals that cannot be written as $\frac{a}{b}$
- Absolute value: $|a|$ is the distance from a to 0
- Percent error: $\frac{|\text{approximate} - \text{exact}|}{|\text{exact}|} * 100\%$

Exponents and Radicals

- $a^m \cdot a^n = a^{m+n}$
- $\frac{a^m}{a^n} = a^{m-n}$, $a \neq 0$
- $(a^m)^n = a^{mn}$
- $(ab)^n = a^n b^n$
- $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$, $b \neq 0$
- $a^0 = 1$, $a \neq 0$
- $a^{-n} = \frac{1}{a^n}$, $a \neq 0$
- $a^{1/n} = \sqrt[n]{a}$
- $a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$
- $\sqrt{ab} = \sqrt{a} \sqrt{b}$
- $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$, $b > 0$

Linear Equations and Inequalities

- Slope: $m = \frac{y_2 - y_1}{x_2 - x_1}$
- Slope-intercept form: $y = mx + b$
- Point-slope form: $y - y_1 = m(x - x_1)$
- Standard form: $Ax + By = C$
- Horizontal line: $y = c$
- Vertical line: $x = c$
- Inequality rule: multiplying or dividing by a negative number reverses the inequality sign.
- Absolute value equations:
 - $|x| = a \rightarrow x = a$ or $x = -a$, if $a \geq 0$
 - $|x| < a \rightarrow -a < x < a$
 - $|x| > a \rightarrow x < -a$ or $x > a$

Systems

- Substitution: solve one equation for a variable and substitute.
- Elimination: add or subtract equations to eliminate a variable.
- One solution: lines intersect once.
- No solution: parallel lines.
- Infinitely many solutions: same line.

Functions

- Function notation: $f(x)$ means the output when input is x.

- Domain: set of allowed inputs.
- Range: set of possible outputs.
- Average rate of change: $\frac{f(b)-f(a)}{b-a}$
- Transformation forms:
 - $f(x)+k$: vertical shift
 - $f(x+k)$: horizontal shift left k
 - $kf(x)$: vertical stretch/compression/reflection
 - $f(kx)$: horizontal compression/stretch/reflection

Polynomials and Factoring

- Standard form: terms in descending powers.
- Degree: greatest exponent of the variable.
- Distributive property: $a(b+c)=ab+ac$
- Difference of squares: $a^2-b^2=(a-b)(a+b)$
- Perfect square trinomials:
 - $a^2+2ab+b^2=(a+b)^2$
 - $a^2-2ab+b^2=(a-b)^2$
- Factoring x^2+bx+c : find numbers with sum b and product c.
- Factoring ax^2+bx+c : use grouping or another valid method.

Quadratics

- Standard form: $f(x)=ax^2+bx+c$
- Vertex form: $f(x)=a(x-h)^2+k$
- Factored form: $f(x)=a(x-r_1)(x-r_2)$
- Axis of symmetry: $x=-\frac{b}{2a}$
- Vertex: $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$
- Quadratic formula: $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$
- Discriminant: $D=b^2-4ac$
 - $D>0$: two real solutions
 - $D=0$: one real solution
 - $D<0$: two complex solutions
- Completing the square: $x^2+bx=(x+\frac{b}{2})^2-(\frac{b}{2})^2$

Exponential Functions and Sequences

- Exponential form: $f(x)=ab^x$
- Growth: $b>1$
- Decay: $0<b<1$
- Arithmetic sequence: $a_n=a_1+(n-1)d$
- Geometric sequence: $a_n=a_1r^{n-1}$
- Recursive arithmetic: $a_n=a_{n-1}+d$
- Recursive geometric: $a_n=ra_{n-1}$

Statistics

- Mean: $\bar{x}=\frac{\text{sum of values}}{\text{number of values}}$
- Median: middle value after ordering.

- Range: maximum minus minimum.
- Interquartile range: $IQR = Q_3 - Q_1$
- Mean absolute deviation: average distance from the mean.
- Residual: observed y-value minus predicted y-value.
- Correlation coefficient r : measures strength and direction of linear association.

UNIT 1: REAL NUMBERS, PROPERTIES, AND PRECISION

Learning Targets

- Classify numbers as rational, irrational, integer, whole, or natural.
- Use properties of rational and irrational numbers.
- Apply order of operations and algebraic properties.
- Round values and choose appropriate precision.

Unit Formula Sheet

- Rational + rational = rational
- Rational * rational = rational
- Rational + irrational = irrational
- Nonzero rational * irrational = irrational
- Commutative: $a+b=b+a$, $ab=ba$
- Associative: $(a+b)+c=a+(b+c)$, $(ab)c=a(bc)$
- Distributive: $a(b+c)=ab+ac$

Sample Problems

Sample 1

Classify $-\frac{8}{4}$.

$$-\frac{8}{4} = -2$$

It is rational and an integer.

Sample 2

Simplify $3[8-(2+1)^2]+5$.

$$(2+1)^2 = 9$$

$$8-9 = -1$$

$$3(-1)+5 = 2$$

Answer: 2

Sample 3

State whether $5+\sqrt{7}$ is rational or irrational.

A rational number plus an irrational number is irrational.

Answer: irrational

Classwork

1. Classify $\sqrt{25}$ as rational or irrational.
2. Classify $\sqrt{18}$ as rational or irrational.
3. Simplify $4(3+2)-7$.
4. Simplify $2^3+6\div3$.
5. Simplify $-3(4-9)+2$.
6. Simplify $\frac{3}{4}+\frac{5}{6}$.
7. Simplify $\frac{7}{8}-\frac{1}{3}$.
8. Simplify $-5+|-12|$.
9. Determine whether $\sqrt{2}\cdot 6$ is rational or irrational.
10. Round 18.7462 to the nearest hundredth.

Homework

1. Classify -9 as natural, whole, integer, rational, or irrational.
2. Classify 0 as whole, integer, rational, or irrational.
3. Simplify $6^2-4(5)$.
4. Simplify $18\div3+2^4$.
5. Simplify $-2[7-(3^2)]$.
6. Simplify $\frac{2}{5}+\frac{7}{10}$.
7. Simplify $\frac{11}{12}-\frac{1}{8}$.
8. Simplify $|-15|-|-6|$.
9. Determine whether $\sqrt{5}+8$ is rational or irrational.
10. Round 3.14159 to the nearest thousandth.

Teacher Answer Key

Classwork Answers

1. rational
2. irrational
3. 13
4. 10
5. 17
6. $\frac{19}{12}$
7. $\frac{13}{24}$

8. 7
9. irrational
10. 18.75

Homework Answers

1. integer and rational
2. whole, integer, rational
3. 16
4. 22
5. 4
6. $\frac{11}{10}$
7. $\frac{19}{24}$
8. 9
9. irrational
10. 3.142

UNIT 2: EXPONENTS, RADICALS, AND RATIONAL EXPONENTS

Learning Targets

- Apply exponent properties.
- Rewrite radicals using rational exponents.
- Rewrite rational exponents using radicals.
- Simplify radicals and expressions with rational exponents.

Unit Formula Sheet

- $a^{\frac{m}{n}} = \sqrt[n]{a^m}$
- $\sqrt[n]{a} = a^{\frac{1}{n}}$
- $a^{-n} = \frac{1}{a^n}$
- $a^0 = 1, a \neq 0$
- $\sqrt{a^2} = |a|$

Sample Problems

Sample 1

Simplify $x^4 \cdot x^7$.

$$x^4 \cdot x^7 = x^{11}$$

Sample 2

Rewrite $\sqrt[3]{y^5}$ using rational exponents.

$$\sqrt[3]{y^5} = y^{5/3}$$

Sample 3

Simplify $16^{3/4}$.

$$16^{3/4} = (\sqrt[4]{16})^3 = 2^3 = 8$$

Classwork

1. Simplify x^3x^8 .
2. Simplify $\frac{a^{10}}{a^4}$.
3. Simplify $(m^2)^5$.
4. Simplify $(3x^2)(4x^5)$.
5. Simplify y^{-3} .
6. Simplify $9^{1/2}$.
7. Simplify $27^{1/3}$.
8. Simplify $81^{3/4}$.
9. Rewrite $\sqrt{x^7}$ using rational exponents.
10. Rewrite $p^{5/2}$ using radicals.

Homework

1. Simplify b^6b^9 .
2. Simplify $\frac{r^{12}}{r^5}$.
3. Simplify $(n^4)^3$.
4. Simplify $(5q^3)(2q^4)$.
5. Simplify z^{-5} .
6. Simplify $25^{1/2}$.
7. Simplify $64^{1/3}$.
8. Simplify $32^{2/5}$.
9. Rewrite $\sqrt[4]{a^9}$ using rational exponents.
10. Rewrite $t^{7/3}$ using radicals.

Teacher Answer Key

Classwork Answers

1. x^{11}
2. a^6
3. m^{10}
4. $12x^7$

5. $\frac{1}{y^3}$
6. 3
7. 3
8. 27
9. $x^{\frac{7}{2}}$
10. $\sqrt{p^5}$ or $p^2\sqrt{p}$

Homework Answers

1. b^{15}
2. r^7
3. n^{12}
4. $10q^7$
5. $\frac{1}{z^5}$
6. 5
7. 4
8. 4
9. $a^{\frac{9}{4}}$
10. $\sqrt[3]{t^7}$

UNIT 3: EXPRESSIONS, STRUCTURE, AND EQUIVALENT FORMS

Learning Targets

- Identify terms, coefficients, constants, and factors.
- Combine like terms.
- Use distribution and factoring to create equivalent expressions.
- Rewrite expressions to reveal structure.

Unit Formula Sheet

- Like terms have identical variable parts.
- Coefficient: numerical factor multiplying a variable term.
- Constant: term with no variable.
- Factor common factor: $ab+ac=a(b+c)$
- Standard linear expression: $ax+b$

Sample Problems

Sample 1

Simplify $4x+7-2x+9$.

$$4x-2x+7+9=2x+16$$

Sample 2

Factor $12x+18$.

$$12x+18=6(2x+3)$$

Sample 3

Rewrite $3(x+4)+2(x-5)$ in standard form.

$$3x+12+2x-10=5x+2$$

Classwork

1. Simplify $7x+3x$.
2. Simplify $9a-4a+6$.
3. Simplify $5(2x+3)$.
4. Simplify $-2(4y-7)$.
5. Simplify $3(x-2)+4(x+5)$.
6. Simplify $8m+2-5m+11$.
7. Factor $10x+15$.
8. Factor $14a-21$.
9. Factor $6x^2+9x$.
10. Identify the coefficient of $-8p^3$.

Homework

1. Simplify $12y-5y$.
2. Simplify $8b+9-3b$.
3. Simplify $7(3x-1)$.
4. Simplify $-5(2n+6)$.
5. Simplify $6(a+1)-2(a-9)$.
6. Simplify $11q-4+2q+10$.
7. Factor $18x+24$.
8. Factor $20m-35$.
9. Factor $12r^2-8r$.
10. Identify the constant in $9x^2-4x+11$.

Teacher Answer Key

Classwork Answers

1. $10x$

2. $5a+6$
3. $10x+15$
4. $-8y+14$
5. $7x+14$
6. $3m+13$
7. $5(2x+3)$
8. $7(2a-3)$
9. $3x(2x+3)$
10. -8

Homework Answers

1. $7y$
2. $5b+9$
3. $21x-7$
4. $-10n-30$
5. $4a+24$
6. $13q+6$
7. $6(3x+4)$
8. $5(4m-7)$
9. $4r(3r-2)$
10. 11

UNIT 4: LINEAR EQUATIONS, INEQUALITIES, AND ABSOLUTE VALUE

Learning Targets

- Solve one-step, two-step, and multi-step linear equations.
- Solve equations with variables on both sides.
- Solve and graph linear inequalities.
- Solve absolute value equations and inequalities.
- Rearrange literal equations.

Unit Formula Sheet

- Equation solving goal: isolate the variable.
- Use inverse operations.
- Keep equations balanced by doing the same operation to both sides.
- Inequality reversal: if multiplying or dividing by a negative, reverse $<$, $>$, $\leq$, $\geq$.
- $|ax+b|=c \rightarrow ax+b=c$ or $ax+b=-c$, if $c \geq 0$.

Sample Problems

Sample 1

Solve $3x+5=20$.

$$3x=15$$

$$x=5$$

Sample 2

Solve $-2x+7 \leq 15$.

$$-2x \leq 8$$

$$x \geq -4$$

Sample 3

Solve $|x-3|=8$.

$$x-3=8 \quad \text{or} \quad x-3=-8$$

$$x=11 \quad \text{or} \quad x=-5$$

Classwork

1. Solve $x+8=15$.
2. Solve $4x=28$.
3. Solve $3x-7=11$.
4. Solve $5(x+2)=35$.
5. Solve $7x-4=3x+12$.
6. Solve $\frac{x}{3}+5=9$.
7. Solve $2x-1 < 9$.
8. Solve $-3x+6 \geq 15$.
9. Solve $|x+4|=10$.
10. Solve $A=lw$ for w .

Homework

1. Solve $x-9=14$.
2. Solve $-6x=42$.
3. Solve $2x+13=31$.
4. Solve $-4(x-3)=20$.
5. Solve $9x+1=4x+26$.
6. Solve $\frac{x}{5}-2=6$.
7. Solve $5x+3 \leq 18$.
8. Solve $-2x-7 > 5$.
9. Solve $|2x-1|=9$.
10. Solve $P=2l+2w$ for l .

Teacher Answer Key

Classwork Answers

1. $x=7$
2. $x=7$
3. $x=6$
4. $x=5$
5. $x=4$
6. $x=12$
7. $x<5$
8. $x\leq-3$
9. $x=6$ or $x=-14$
10. $w=\frac{A}{l}$

Homework Answers

1. $x=23$
2. $x=-7$
3. $x=9$
4. $x=-2$
5. $x=5$
6. $x=40$
7. $x\leq 3$
8. $x<-6$
9. $x=5$ or $x=-4$
10. $l=\frac{P-2w}{2}$

UNIT 5: COORDINATE GRAPHING AND LINEAR FUNCTIONS

Learning Targets

- Find slope from points, tables, and equations.
- Graph lines in slope-intercept, point-slope, and standard form.
- Write equations of lines.
- Identify parallel and perpendicular slopes.

Unit Formula Sheet

- $m=\frac{y_2-y_1}{x_2-x_1}$
- $y=mx+b$
- $y-y_1=m(x-x_1)$
- Parallel lines have equal slopes.
- Perpendicular lines have slopes whose product is -1 .
- x-intercept: set $y=0$.

- y-intercept: set $x=0$.

Sample Problems

Sample 1

Find the slope through (2,5) and (6,13).

$$m = \frac{13-5}{6-2} = \frac{8}{4} = 2$$

Sample 2

Write the equation with slope 3 and y-intercept -4.

$$y = 3x - 4$$

Sample 3

Find the x-intercept of $2x + 3y = 12$.

Set $y=0$:

$$2x = 12$$

$$x = 6$$

Answer: (6,0)

Classwork

1. Find the slope through (1,2) and (4,8).
2. Find the slope through (-2,5) and (3,-5).
3. Write the equation with slope 4 and y-intercept 1.
4. Write the equation with slope -2 and y-intercept 7.
5. Find the slope of $y = \frac{3}{2}x - 9$.
6. Find the y-intercept of $y = -5x + 4$.
7. Find the x-intercept of $3x + 2y = 18$.
8. Find the y-intercept of $4x - 5y = 20$.
9. Write a line parallel to $y = 2x - 6$ through (0,5).
10. Write a line perpendicular to $y = \frac{1}{2}x + 3$ through (0,-1).

Homework

1. Find the slope through (0,4) and (2,10).
2. Find the slope through (-3,-1) and (1,7).
3. Write the equation with slope $\frac{1}{2}$ and y-intercept -3.
4. Write the equation with slope -4 and y-intercept 6.

5. Find the slope of $y = -\frac{4}{5}x + 2$.
6. Find the y-intercept of $y = 7x - 8$.
7. Find the x-intercept of $5x + y = 30$.
8. Find the y-intercept of $2x + 3y = 15$.
9. Write a line parallel to $y = -3x + 1$ through $(0, 4)$.
10. Write a line perpendicular to $y = -\frac{1}{4}x + 2$ through $(0, 9)$.

Teacher Answer Key

Classwork Answers

1. 2
2. -2
3. $y = 4x + 1$
4. $y = -2x + 7$
5. $\frac{3}{2}$
6. $(0, 4)$
7. $(6, 0)$
8. $(0, -4)$
9. $y = 2x + 5$
10. $y = -2x - 1$

Homework Answers

1. 3
2. 2
3. $y = \frac{1}{2}x - 3$
4. $y = -4x + 6$
5. $-\frac{4}{5}$
6. $(0, -8)$
7. $(6, 0)$
8. $(0, 5)$
9. $y = -3x + 4$
10. $y = 4x + 9$

UNIT 6: SYSTEMS OF EQUATIONS AND INEQUALITIES

Learning Targets

- Solve linear systems by graphing, substitution, and elimination.
- Classify systems as one solution, no solution, or infinitely many solutions.
- Solve systems involving a line and a quadratic.
- Solve and graph systems of linear inequalities.

Unit Formula Sheet

- Solution of a system: ordered pair satisfying every equation.
- Elimination works because adding a multiple of one equation to another preserves the solution set.
- Linear-quadratic systems: substitute the linear expression into the quadratic equation.

Sample Problems

Sample 1

Solve:

$$y = x + 2$$

$$y = 3x - 4$$

Set equal:

$$x + 2 = 3x - 4$$

$$6 = 2x$$

$$x = 3$$

$$y = 5$$

Answer: (3,5)

Sample 2

Solve:

$$2x + y = 7$$

$$3x - y = 8$$

Add:

$$5x = 15$$

$$x = 3$$

$$2(3) + y = 7 \rightarrow y = 1$$

Answer: (3,1)

Classwork

1. Solve: $y = x + 1$, $y = 2x - 3$.
2. Solve: $y = -x + 5$, $y = x + 1$.
3. Solve: $x + y = 9$, $x - y = 1$.
4. Solve: $2x + y = 10$, $x - y = 2$.
5. Solve: $3x + 2y = 16$, $x + 2y = 8$.
6. Classify: $y = 2x + 1$, $y = 2x - 4$.
7. Classify: $2x + y = 5$, $4x + 2y = 10$.
8. Solve: $y = x^2$, $y = 4$.

9. Solve: $y=x^2+1$, $y=5$.
10. State one solution to the inequality system $y>x$, $y<5$.

Homework

1. Solve: $y=x+4$, $y=3x$.
2. Solve: $y=-2x+6$, $y=x-3$.
3. Solve: $x+y=12$, $x-y=4$.
4. Solve: $4x+y=18$, $x-y=3$.
5. Solve: $5x+y=19$, $2x-y=2$.
6. Classify: $y=-x+2$, $y=-x+7$.
7. Classify: $3x-y=6$, $6x-2y=12$.
8. Solve: $y=x^2$, $y=9$.
9. Solve: $y=x^2-2$, $y=7$.
10. State one solution to the inequality system $y<2x$, $y>0$.

Teacher Answer Key

Classwork Answers

1. (4,5)
2. (2,3)
3. (5,4)
4. (4,2)
5. (4,2)
6. no solution
7. infinitely many solutions
8. (-2,4), (2,4)
9. (-2,5), (2,5)
10. Sample: (1,2)

Homework Answers

1. (2,6)
2. (3,0)
3. (8,4)
4. $(\frac{21}{5}, \frac{65}{5})$
5. (3,4)
6. no solution
7. infinitely many solutions
8. (-3,9), (3,9)
9. (-3,7), (3,7)
10. Sample: (1,1)

Learning Targets

- Determine whether relations are functions.
- Evaluate function notation.
- Find domain and range from sets, tables, and equations.
- Calculate average rate of change.
- Identify transformations of functions.
- Find inverse functions for simple linear functions.

Unit Formula Sheet

- Function rule: each input has exactly one output.
- $f(a)$: substitute a for x .
- Average rate of change: $\frac{f(b)-f(a)}{b-a}$
- Inverse of a linear function: replace $f(x)$ with y , switch x and y , solve for y .

Sample Problems

Sample 1

If $f(x)=2x^2-3$, find $f(4)$.

$$f(4)=2(4)^2-3=32-3=29$$

Sample 2

Find the inverse of $f(x)=3x-5$.

$$y=3x-5$$

$$x=3y-5$$

$$x+5=3y$$

$$y=\frac{x+5}{3}$$

$$\text{Answer: } f^{-1}(x)=\frac{x+5}{3}$$

Classwork

1. If $f(x)=3x+2$, find $f(5)$.
2. If $g(x)=x^2-4$, find $g(-3)$.
3. If $h(x)=\frac{x}{2}+7$, find $h(10)$.
4. Find the domain of $\{(-2,5),(0,1),(4,9)\}$.
5. Find the range of $\{(-2,5),(0,1),(4,9)\}$.
6. Is $\{(1,2),(1,3),(4,5)\}$ a function
7. Find the average rate of change of $f(x)=x^2$ from $x=1$ to $x=4$.

8. Describe the transformation $f(x)+6$.
9. Describe the transformation $f(x-3)$.
10. Find the inverse of $f(x)=2x+8$.

Homework

1. If $f(x)=4x-1$, find $f(6)$.
2. If $g(x)=x^2+2x$, find $g(3)$.
3. If $h(x)=\frac{x-4}{3}$, find $h(13)$.
4. Find the domain of $\{(-1,7),(2,8),(5,8)\}$.
5. Find the range of $\{(-1,7),(2,8),(5,8)\}$.
6. Is $\{(2,4),(3,4),(4,4)\}$ a function
7. Find the average rate of change of $f(x)=2x+1$ from $x=-2$ to $x=3$.
8. Describe the transformation $f(x)-4$.
9. Describe the transformation $2f(x)$.
10. Find the inverse of $f(x)=5x-10$.

Teacher Answer Key

Classwork Answers

1. 17
2. 5
3. 12
4. $\{-2,0,4\}$
5. $\{1,5,9\}$
6. no
7. 5
8. vertical shift up 6
9. horizontal shift right 3
10. $f^{-1}(x)=\frac{x+8}{2}$

Homework Answers

1. 23
2. 15
3. 3
4. $\{-1,2,5\}$
5. $\{7,8\}$
6. yes
7. 2
8. vertical shift down 4
9. vertical stretch by factor 2
10. $f^{-1}(x)=\frac{x+10}{5}$

Learning Targets

- Add, subtract, and multiply polynomials.
- Classify polynomials by degree and number of terms.
- Factor common factors, trinomials, and special products.
- Use factoring to reveal zeros.

Unit Formula Sheet

- Monomial: one term.
- Binomial: two terms.
- Trinomial: three terms.
- Degree of polynomial: greatest exponent after simplification.
- FOIL: $(a+b)(c+d)=ac+ad+bc+bd$
- Zero product property: if $ab=0$, then $a=0$ or $b=0$.

Sample Problems

Sample 1

Add $(3x^2+2x-5)+(x^2-7x+9)$.

$$4x^2-5x+4$$

Sample 2

Multiply $(x+4)(x-6)$.

$$x^2-6x+4x-24=x^2-2x-24$$

Sample 3

Factor $x^2+7x+12$.

Numbers with product 12 and sum 7: 3 and 4.

$$x^2+7x+12=(x+3)(x+4)$$

Classwork

1. Add $(2x^2+3x+1)+(5x^2-x+4)$.

2. Subtract $(7x^2+2x-8)-(3x^2+5x-1)$.
3. Multiply $3x(2x^2-4x+5)$.
4. Multiply $(x+2)(x+5)$.
5. Multiply $(x-3)(x+7)$.
6. Multiply $(2x+1)(x-4)$.
7. Factor $x^2+9x+20$.
8. Factor x^2-5x+6 .
9. Factor x^2-16 .
10. Factor $2x^2+7x+3$.

Homework

1. Add $(4x^2-2x+6)+(x^2+8x-9)$.
2. Subtract $(9x^2-x+2)-(4x^2+3x+7)$.
3. Multiply $-2x(5x^2+x-3)$.
4. Multiply $(x+6)(x+1)$.
5. Multiply $(x-8)(x+2)$.
6. Multiply $(3x-2)(x+5)$.
7. Factor $x^2+8x+15$.
8. Factor $x^2-7x+10$.
9. Factor x^2-25 .
10. Factor $3x^2+10x+3$.

Teacher Answer Key

Classwork Answers

1. $7x^2+2x+5$
2. $4x^2-3x-7$
3. $6x^3-12x^2+15x$
4. $x^2+7x+10$
5. $x^2+4x-21$
6. $2x^2-7x-4$
7. $(x+4)(x+5)$
8. $(x-2)(x-3)$
9. $(x-4)(x+4)$
10. $(2x+1)(x+3)$

Homework Answers

1. $5x^2+6x-3$
2. $5x^2-4x-5$
3. $-10x^3-2x^2+6x$
4. x^2+7x+6
5. $x^2-6x-16$
6. $3x^2+13x-10$
7. $(x+3)(x+5)$
8. $(x-5)(x-2)$
9. $(x-5)(x+5)$

10. $(3x+1)(x+3)$

UNIT 9: QUADRATIC FUNCTIONS AND EQUATIONS

Learning Targets

- Graph quadratic functions and identify key features.
- Solve quadratic equations by factoring, square roots, completing the square, and quadratic formula.
- Use the discriminant to classify solutions.
- Rewrite quadratics in equivalent forms.

Unit Formula Sheet

- $f(x)=ax^2+bx+c$
- Axis: $x=-\frac{b}{2a}$
- Vertex: substitute the axis value into the function.
- Factored zeros: $a(x-r_1)(x-r_2)=0 \Rightarrow x=r_1, r_2$
- Quadratic formula: $x=\frac{-b \pm \sqrt{b^2-4ac}}{2a}$
- Complete the square: add $(\frac{b}{2})^2$ to both sides after isolating x^2+bx .

Sample Problems

Sample 1

Solve $x^2-9=0$.

$$x^2=9$$
$$x=\pm 3$$

Sample 2

Solve $x^2+5x+6=0$.

$$(x+2)(x+3)=0$$
$$x=-2, -3$$

Sample 3

Find the vertex of $f(x)=x^2-4x+1$.

$$x = -\frac{-4 \pm \sqrt{2(1)}}{2} = 2$$

$$f(2) = 4 - 8 + 1 = -3$$

Vertex: (2, -3)

Classwork

1. Solve $x^2 = 49$.
2. Solve $x^2 - 25 = 0$.
3. Solve $x^2 + 6x + 8 = 0$.
4. Solve $x^2 - 3x - 10 = 0$.
5. Solve $2x^2 + 5x + 2 = 0$.
6. Find the axis of $y = x^2 - 8x + 3$.
7. Find the vertex of $y = x^2 + 2x - 5$.
8. Find the discriminant of $x^2 - 4x + 9 = 0$.
9. Solve $x^2 + 4x + 1 = 0$ using the quadratic formula.
10. Complete the square: rewrite $x^2 + 10x$ as a squared binomial minus a constant.

Homework

1. Solve $x^2 = 81$.
2. Solve $x^2 - 36 = 0$.
3. Solve $x^2 + 7x + 12 = 0$.
4. Solve $x^2 - x - 20 = 0$.
5. Solve $3x^2 + 11x + 6 = 0$.
6. Find the axis of $y = x^2 + 10x - 2$.
7. Find the vertex of $y = x^2 - 6x + 4$.
8. Find the discriminant of $2x^2 + 3x - 5 = 0$.
9. Solve $x^2 - 2x - 3 = 0$ using the quadratic formula.
10. Complete the square: rewrite $x^2 - 12x$ as a squared binomial minus a constant.

Teacher Answer Key

Classwork Answers

1. $x = \pm 7$
2. $x = \pm 5$
3. $x = -2, -4$
4. $x = 5, -2$
5. $x = -\frac{5}{2}, -2$
6. $x = 4$
7. $(-1, -6)$
8. -20
9. $x = -2 \pm \sqrt{3}$
10. $(x+5)^2 - 25$

Homework Answers

1. $x = \pm 9$
2. $x = \pm 6$
3. $x = -3, -4$
4. $x = 5, -4$
5. $x = -\frac{23}{3}, -3$
6. $x = -5$
7. $(3, -5)$
8. 49
9. $x = 3, -1$
10. $(x-6)^2 - 36$

UNIT 10: EXPONENTIAL FUNCTIONS AND SEQUENCES

Learning Targets

- Identify exponential growth and decay.
- Evaluate exponential functions.
- Rewrite exponential expressions to show rates.
- Write arithmetic and geometric sequences explicitly and recursively.
- Compare linear, quadratic, and exponential patterns from tables.

Unit Formula Sheet

- Exponential function: $f(x) = ab^x$
- Growth if $b > 1$; decay if $0 < b < 1$
- Arithmetic explicit: $a_n = a_1 + (n-1)d$
- Geometric explicit: $a_n = a_1 r^{n-1}$
- Arithmetic recursive: $a_1 = \text{first term}, a_n = a_{n-1} + d$
- Geometric recursive: $a_1 = \text{first term}, a_n = r a_{n-1}$

Sample Problems

Sample 1

Evaluate $f(x) = 3(2)^x$ when $x = 4$.

$$f(4) = 3(2)^4 = 48$$

Sample 2

Find an explicit formula for 5, 8, 11, 14,

$$a_1=5, \text{quad } d=3$$

$$a_n=5+3(n-1)$$

Sample 3

Find an explicit formula for 2,10,50,250,....

$$a_1=2, \text{quad } r=5$$

$$a_n=2(5)^{n-1}$$

Classwork

1. Evaluate 2^5 .
2. Evaluate $4(3)^2$.
3. Determine growth or decay: $y=5(1.2)^x$.
4. Determine growth or decay: $y=7(0.8)^x$.
5. Find the next three terms: 4,9,14,19,....
6. Find the next three terms: 3,6,12,24,....
7. Write an explicit formula for 7,10,13,16,....
8. Write an explicit formula for 5,15,45,135,....
9. Write a recursive rule for 2,7,12,17,....
10. Write a recursive rule for 4,20,100,500,....

Homework

1. Evaluate 3^4 .
2. Evaluate $2(5)^3$.
3. Determine growth or decay: $y=9(1.05)^x$.
4. Determine growth or decay: $y=12(0.6)^x$.
5. Find the next three terms: 1,5,9,13,....
6. Find the next three terms: 2,8,32,128,....
7. Write an explicit formula for 6,11,16,21,....
8. Write an explicit formula for 3,12,48,192,....
9. Write a recursive rule for 9,12,15,18,....
10. Write a recursive rule for 6,30,150,750,....

Teacher Answer Key

Classwork Answers

1. 32
2. 36
3. growth
4. decay
5. 24,29,34
6. 48,96,192
7. $a_n=7+3(n-1)$

8. $a_n = 5(3)^{n-1}$
9. $a_1 = 2, a_n = a_{n-1} + 5$
10. $a_1 = 4, a_n = 5a_{n-1}$

Homework Answers

1. 81
2. 250
3. growth
4. decay
5. 17, 21, 25
6. 512, 2048, 8192
7. $a_n = 6 + 5(n-1)$
8. $a_n = 3(4)^{n-1}$
9. $a_1 = 9, a_n = a_{n-1} + 3$
10. $a_1 = 6, a_n = 5a_{n-1}$

UNIT 11: RATIONAL EXPRESSIONS, RADICALS, AND INVERSES

Learning Targets

- Simplify rational expressions.
- Multiply and divide rational expressions.
- Add and subtract rational expressions with common denominators.
- Simplify radical expressions.
- Find inverses of linear and simple quadratic-root functions.

Unit Formula Sheet

- Rational expression restrictions: denominator $\neq 0$.
- $\frac{ab}{ac} = \frac{b}{c}, a \neq 0, c \neq 0$
- $\frac{ab}{c} \cdot \frac{cd}{e} = \frac{ac}{e} \cdot \frac{bd}{c}$
- $\frac{ab}{c} \div \frac{cd}{e} = \frac{ab}{c} \cdot \frac{e}{cd}$
- $\frac{ab}{c} + \frac{cb}{c} = \frac{a+c}{b}$
- $\sqrt{ab} = \sqrt{a} \sqrt{b}$
- Rationalizing: $\frac{1}{\sqrt{a}} = \frac{\sqrt{a}}{a}$

Sample Problems

Sample 1

Simplify $\frac{x^2 + 5x}{x}$.

$$\frac{x(x+5)}{x} = x+5, \text{quad } x \neq 0$$

Sample 2

Simplify $\sqrt{72}$.

$$\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$$

Sample 3

Find the inverse of $f(x) = \sqrt{x-4}$.

$$y = \sqrt{x-4}$$

$$x = \sqrt{y-4}$$

$$x^2 = y-4$$

$$y = x^2 + 4$$

Answer: $f^{-1}(x) = x^2 + 4$, with inverse domain $x \geq 0$.

Classwork

1. Simplify $\frac{6x}{3}$.
2. Simplify $\frac{x^2+3x}{x}$.
3. Simplify $\frac{x^2-9}{x-3}$.
4. Multiply $\frac{2x}{5} \cdot \frac{15}{4x}$.
5. Divide $\frac{x}{3} \div \frac{2x}{9}$.
6. Add $\frac{3}{x} + \frac{5}{x}$.
7. Simplify $\sqrt{50}$.
8. Simplify $\sqrt{27}$.
9. Rationalize $\frac{4}{\sqrt{2}}$.
10. Find the inverse of $f(x) = x-9$.

Homework

1. Simplify $\frac{10x}{5}$.
2. Simplify $\frac{x^2-4x}{x}$.
3. Simplify $\frac{x^2-16}{x-4}$.
4. Multiply $\frac{3x}{7} \cdot \frac{14}{9x}$.
5. Divide $\frac{2x}{5} \div \frac{x}{10}$.
6. Add $\frac{7}{x} + \frac{2}{x}$.
7. Simplify $\sqrt{98}$.
8. Simplify $\sqrt{48}$.
9. Rationalize $\frac{3}{\sqrt{3}}$.
10. Find the inverse of $f(x) = x+11$.

Teacher Answer Key

Classwork Answers

1. $2x$
2. $x+3$, $x \neq 0$
3. $x+3$, $x \neq 3$
4. $\frac{3}{2}$
5. $\frac{3}{2}$
6. $\frac{8}{x}$
7. $5\sqrt{2}$
8. $3\sqrt{3}$
9. $2\sqrt{2}$
10. $f^{-1}(x) = x+9$

Homework Answers

1. $2x$
2. $x-4$, $x \neq 0$
3. $x+4$, $x \neq 4$
4. $\frac{2}{3}$
5. 4
6. $\frac{9}{x}$
7. $7\sqrt{2}$
8. $4\sqrt{3}$
9. $\sqrt{3}$
10. $f^{-1}(x) = x-11$

UNIT 12: STATISTICS, DATA DISPLAYS, AND LINEAR MODELS

Learning Targets

- Calculate measures of center and spread.
- Interpret dot plots, histograms, and box plot values computationally.
- Build and use two-way frequency tables.
- Calculate relative frequencies.
- Use linear models, residuals, and correlation values.

Unit Formula Sheet

- Mean: $\bar{x} = \frac{\sum x}{n}$
- Median: middle number in ordered list.
- Mode: most frequent value.
- Range: $\max - \min$
- IQR: $Q_3 - Q_1$
- Residual: $y - \hat{y}$
- Linear model prediction: substitute x into $\hat{y} = mx + b$

Sample Problems

Sample 1

Find the mean of 4,6,8,10.

$$\bar{x} = \frac{4+6+8+10}{4} = 7$$

Sample 2

Find the median of 3,9,4,10,6.

Order: 3,4,6,9,10. Median: 6

Sample 3

For $\hat{y} = 2x + 1$, find the residual when $x = 4$ and observed $y = 12$.

Predicted:

$$\hat{y} = 2(4) + 1 = 9$$

Residual:

$$12 - 9 = 3$$

Classwork

1. Find the mean of 2,4,6,8.
2. Find the median of 7,1,9,3,5.
3. Find the mode of 2,3,3,4,5,5,5.
4. Find the range of 12,4,18,9.
5. Find Q_1 , Q_3 , and IQR for 2,4,6,8,10,12,14.
6. For $\hat{y} = 3x - 2$, predict y when $x = 5$.
7. For $\hat{y} = 4x + 1$, find the residual when $x = 2$ and observed $y = 12$.
8. Compute the joint relative frequency: table total 40, cell count 10.
9. Compute the marginal relative frequency: table total 50, row total 20.
10. State whether $r = 0.92$ shows positive or negative correlation.

Homework

1. Find the mean of 5,10,15,20.
2. Find the median of 8,2,12,4,10,6.
3. Find the mode of 1,1,2,3,3,3,4.
4. Find the range of 21,15,7,30.
5. Find Q_1 , Q_3 , and IQR for 1,3,5,7,9,11,13.
6. For $\hat{y} = 2x + 6$, predict y when $x = 7$.

7. For $y=5x-3$, find the residual when $x=4$ and observed $y=20$.
8. Compute the joint relative frequency: table total 80, cell count 12.
9. Compute the marginal relative frequency: table total 60, column total 15.
10. State whether $r=-0.81$ shows positive or negative correlation.

Teacher Answer Key

Classwork Answers

1. 5
2. 5
3. 5
4. 14
5. $Q_1=4, Q_3=12, IQR=8$
6. 13
7. 3
8. 0.25 or 25%
9. 0.4 or 40%
10. positive

Homework Answers

1. 12.5
2. 7
3. 3
4. 23
5. $Q_1=3, Q_3=11, IQR=8$
6. 20
7. 3
8. 0.15 or 15%
9. 0.25 or 25%
10. negative

UNIT 13: CUMULATIVE REVIEW AND FINAL EXAM PRACTICE

Cumulative Formula Sheet

- Exponents: $a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, $a^{-n} = \frac{1}{a^n}$
- Radicals: $a^{\frac{1}{n}} = \sqrt[n]{a}$
- Slope: $m = \frac{y_2 - y_1}{x_2 - x_1}$
- Line: $y = mx + b$
- Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Axis: $x = -\frac{b}{2a}$
- Average rate of change: $\frac{f(b) - f(a)}{b - a}$
- Arithmetic sequence: $a_n = a_1 + (n-1)d$

- Geometric sequence: $a_n = a_1 r^{n-1}$
- Mean: $\bar{x} = \frac{\sum x}{n}$

Final Exam Practice A: Student Version

1. Simplify $3(2x-5)+4x$.
2. Simplify $\frac{x^9}{x^4}$.
3. Simplify $64^{\frac{2}{3}}$.
4. Solve $5x-7=18$.
5. Solve $-2x+3 \leq 11$.
6. Solve $|x-6|=4$.
7. Find the slope through (2,3) and (8,15).
8. Write the equation of a line with slope -3 and y-intercept 5.
9. Solve the system $y=x+2$, $y=-x+8$.
10. Evaluate $f(x)=x^2-2x$ at $x=5$.
11. Find the average rate of change of $f(x)=3x+1$ from $x=0$ to $x=4$.
12. Multiply $(x+7)(x-2)$.
13. Factor $x^2+2x-15$.
14. Solve $x^2-4x-12=0$.
15. Find the vertex of $y=x^2-10x+9$.
16. Determine growth or decay: $y=4(0.75)^x$.
17. Write an explicit formula for 3,8,13,18,....
18. Write an explicit formula for 2,6,18,54,....
19. Simplify $\sqrt{75}$.
20. Find the mean of 6,8,10,12.

Final Exam Practice B: Student Version

1. Simplify $-4(x+3)+9x$.
2. Simplify $(a^3)^4$.
3. Simplify $125^{\frac{1}{3}}$.
4. Solve $\frac{x}{4}+6=10$.
5. Solve $3x-8>13$.
6. Solve $|2x+1|=7$.
7. Find the slope through (-1,4) and (3,12).
8. Write the equation of a line with slope $\frac{2}{3}$ and y-intercept -1.
9. Solve the system $x+y=11$, $x-y=3$.
10. Evaluate $g(x)=2x^2+1$ at $x=-3$.
11. Find the average rate of change of $f(x)=x^2$ from $x=2$ to $x=6$.
12. Multiply $(x-5)(x-4)$.
13. Factor $x^2-11x+30$.
14. Solve $x^2+9x+20=0$.
15. Find the vertex of $y=x^2+6x-1$.
16. Determine growth or decay: $y=2(1.4)^x$.
17. Write an explicit formula for 10,7,4,1,....
18. Write an explicit formula for 5,20,80,320,....
19. Simplify $\sqrt{108}$.
20. Find the median of 9,1,5,7,3.

Teacher Answer Key

Final Exam Practice A Answers

1. $10x-15$
2. x^5
3. 16
4. $x=5$
5. $x \geq -4$
6. $x=10$ or $x=2$
7. 2
8. $y=-3x+5$
9. (3,5)
10. 15
11. 3
12. $x^2+5x-14$
13. $(x+5)(x-3)$
14. $x=6, -2$
15. (5, -16)
16. decay
17. $a_n=3+5(n-1)$
18. $a_n=2(3)^{n-1}$
19. $5\sqrt{3}$
20. 9

Final Exam Practice B Answers

1. $5x-12$
2. a^{12}
3. 5
4. $x=16$
5. $x>7$
6. $x=3$ or $x=-4$
7. 2
8. $y=\frac{2}{3}x-1$
9. (7,4)
10. 19
11. 8
12. $x^2-9x+20$
13. $(x-5)(x-6)$
14. $x=-4, -5$
15. (-3, -10)
16. growth
17. $a_n=10-3(n-1)$
18. $a_n=5(4)^{n-1}$
19. $6\sqrt{3}$
20. 5

TEACHER IMPLEMENTATION NOTES

Suggested Weekly Routine

- Day 1: Formula sheet, vocabulary, and sample problems.
- Day 2: Guided classwork set, error analysis, and fluency drills.
- Day 3: Mixed practice and short quiz.
- Day 4: Homework review, reteaching, and extension problems.
- Day 5: Unit checkpoint or cumulative review.

Computational Practice Expectations

Students should show:

- Substitution steps for evaluation problems.
- Inverse operations for equations.
- Sign changes when solving inequalities involving multiplication or division by negatives.
- Factoring steps for quadratic equations.
- Formula substitution for slope, vertex, quadratic formula, statistics, and sequences.

Recommended Assessment Types

- 5-question daily entrance ticket.
- 10-question weekly computational quiz.
- Unit test with 20 to 30 computational items.
- Cumulative final exam using Units 1 through 13.